Day: _____

Date: _____ Pg No 1

No 1 For a real valued function $f(x)$ on $(-\infty, \infty)$ show that

$$\overline{F(\alpha)} = F(-\alpha)$$

Sol

$$F(\alpha) = \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du$$

Taking conjugate we have

$$\overline{F(\alpha)} = \int_{-\infty}^{\infty} f(u) e^{-i\alpha u} du$$

$$\int_{-\infty}^{\infty} \overline{f(u)} \, \overline{(e^{-i\alpha u})} \, du$$

$$\int_{-\infty}^{\infty} f(u) \, e^{i\alpha u} \, du$$

$$\int_{-\infty}^{\infty} f(u) \, e^{-i(-\alpha)u} \, du = F(-\alpha)$$

Hence proved that

$$\overline{F(\alpha)} = F(-\alpha).$$

Date: Pg No 2

Q No 2 Convolution of Commutative

$$f * g = g * f$$

Sol By defination of Convolution

$$F * g = \int_{-\infty}^{\infty} f(u)\, g(x-u)\, du$$

Put

$$x - u = t$$

$$u = x - t$$

$$du = -dt$$

So, we get

$$F * g = \int_{\infty}^{-\infty} f(x-t)\, g(u)\, (-dt)$$

$$\int_{-\infty}^{\infty} g(t)\, f(x-t)\, dt$$

$$g * f$$

So, Proved

$$f * g = g * f$$

Q No 3 Improper integral by Gamma function.

$$\int_0^\infty x^m e^{-ax^n}\, dx$$

Sol

$$I = \int_0^\infty x^m e^{-ax^n}\, dx$$

let $ax^n = u \Rightarrow x^n = u/a$

$\Rightarrow x = (u/a)^{1/n}$

$u = ax^n$

$du = a[nx^{n-1}]dx$

As

$x \to 0 \quad u = 0 \quad x \to \infty, u = \infty$

Now putting in (I)

$\therefore u = ax^n \quad x = [u/a]^{1/n}$

$$\int_0^\infty \left[\left(\frac{u}{a}\right)^{1/n}\right]^m - e^{-u} \cdot d[u/a]^{1/n}$$

After Simplifing

$$I = \frac{1}{a^{m/n}} \left[u^{m/n}\, e^{-u} \left\{ \frac{1}{n} (u/a)^{\frac{1-n}{n}} \cdot \frac{1}{a}\, du \right\} \right.$$

$$I = \frac{1}{a^{\frac{m}{n}+\frac{1-n}{n}+1}} \int_0^\infty \frac{1}{n} u^{\frac{m+1}{n}-1} e^{-u} du$$

$$= \frac{1}{n a^{\frac{m+1}{n}}} \gamma\left[\frac{m+1}{n}\right]$$

This is the required result.

Q No 4 Find the general solution.

$$\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$$

$$u = e^{ax+by}$$

Sol: Taking 1st and 2nd derivatives "x"

$$\frac{\partial u}{\partial x} = e^{ax+by} \, d/dx(ax+by) = a e^{ax+by}$$

$$= au$$

$$\frac{\partial^2 u}{\partial x^2} = a \frac{\partial}{\partial x}(u)$$

$$\frac{\partial^2 u}{\partial x^2} = a \frac{\partial u}{\partial x} \qquad \therefore \frac{\partial u}{\partial x} = au$$

$$\partial^2 u / \partial x^2 = a^2 u$$

Day: Date: Pg No 5

Taking derivatives "y"

$$u = e^{ax+by}$$

$$\partial u/\partial y = e^{ax+by} \cdot b$$

$$= bu$$

$$\partial^2 u/\partial y^2 = b\, \partial u/\partial y$$

$$\partial^2 u/\partial y^2 = b^2 u$$

$$u = e^{ax+by}$$

$$\frac{\partial u}{\partial x} = e^{ax+b} \cdot a$$

$$\frac{\partial^2 u}{\partial x \partial y} = ae^{ax+b}$$

$$= a e^{ax+b} \cdot b$$

$$\partial^2/\partial x \partial y = abu$$

Putting in

$$\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial^2 u}{\partial x \partial y} + 2\frac{\partial^2 u}{\partial y^2} = 0$$

$$a^2 u + 3abu + 2b^2 u = 0$$

$$a^2u + 3abu + 2b^2u = 0$$

$$u(a^2 + 3ab + 2b) = 0$$

$$(a+2b)(a+b) = 0$$

$$a = -2b \quad , \quad a = -b$$

2 NOS

$$\Gamma\left(-\tfrac{5}{2}\right)$$

We Know That

$$\Gamma(n) = \frac{\Gamma(n+1)}{n}$$

$$n = -\tfrac{5}{2}$$

$$\Gamma\left(-\tfrac{5}{2}\right) = \frac{\Gamma\left(-\tfrac{5}{2}+1\right)}{-\tfrac{5}{2}}$$

$$= \frac{\Gamma\left(-\tfrac{3}{2}\right)}{-\tfrac{5}{2}}$$

$$= -\frac{18\sqrt{\pi}}{15}$$

Day: Date: Pg No 7

Q No 6 Obtaining Normalizing Constants from orthognol sets.

$$\left\{1, \frac{\sin \pi x}{l}, \frac{\cos \pi x}{l}, \frac{\sin 2\pi x}{l}, \frac{\cos 2\pi x}{l} \ldots\ldots\right\}$$

Sol

condition of orthognal

$$\int_{-l}^{l} \phi_m(x)\, \phi_n(x)\, dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

1) Now $\int_{-l}^{l} 1 \cdot \frac{\sin k\pi x}{l}\, dx = 0$

$$\int_{-l}^{l} 1 \cdot \cos \frac{k\pi x}{l}\, dx = 0$$

2) $\int_{-l}^{l} \sin\left(\frac{k\pi x}{l}\right) \cos\left(\frac{p\pi x}{l}\right) \quad k \neq p$

$= 0$

3) $\int_{-l}^{l} \sin\left(\frac{m\pi x}{l}\right) \sin\left(\frac{n\pi x}{l}\right)$

$= l$ $\begin{cases} l & m = n \\ 0 & m \neq n \end{cases}$

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Date: Page 108

$$\int_{-l}^{l} \cos\left(\frac{n\pi x}{l}\right) \cos\left(\frac{m\pi x}{l}\right)$$

$$= l$$

Now

$$\int_{-l}^{l} \left(\frac{1}{\sqrt{l}} \sin\frac{m\pi x}{l}\right)^2 = 1$$

$$\int_{-l}^{l} \left(\frac{1}{\sqrt{l}} \cos\frac{m\pi x}{l}\right)^2 = 1$$

$$\int_{-l}^{l} (1)^2 \, dx \qquad \because (l - (-l)) = 2l$$

$$= 2l$$

$$\int_{-l}^{l} \left(\frac{1}{\sqrt{2l}}\right)^2 dx = 1$$

orthonormalize constant is

$$\frac{1}{\sqrt{2l}}, \frac{1}{\sqrt{l}}$$

Corresponding Constants.

$$\left\{ \frac{1}{\sqrt{2l}}, \frac{1}{\sqrt{l}} \sin\frac{\pi x}{l}, \frac{1}{\sqrt{l}} \cos\frac{\pi x}{l}, \frac{1}{\sqrt{l}} \sin\frac{2\pi x}{l}, \frac{1}{\sqrt{l}} \cos\frac{2\pi x}{l} \right\}$$

QNO7 Check the term by term differentiable of Fourier series

$$F(x) = \frac{4}{\pi}\left[\sin\frac{\pi x}{2} - \frac{1}{2}\sin\frac{2\pi x}{2} + \frac{1}{3}\sin\frac{3\pi x}{2} \right]$$

$$f'(x) = \frac{4}{\pi}\left[\frac{\pi}{2}\cos\frac{\pi x}{2} - \frac{1}{\cancel{2}}\frac{\cancel{2}\pi}{2}\cos\frac{2\pi x}{2} + \frac{1}{\cancel{3}}\cdot\frac{3\pi}{2}\cos\frac{3\pi x}{2} - \cdots \right]$$

$$= \frac{4}{\pi}\left[\frac{\pi}{2}\cos\frac{\pi x}{2} - \frac{\pi}{2}\cos\frac{2\pi x}{2} + \frac{\pi}{2}\cos\frac{3\pi x}{2} - \cdots \right]$$

$$= \frac{4}{\cancel{\pi}}\cdot\frac{\pi}{2}\left[\cos\frac{\pi x}{2} - \cos\frac{2\pi x}{2} + \cos\frac{3\pi x}{2} - \cdots \right]$$

$$2\left[\cos\frac{\pi x}{2} - \cos\frac{2\pi x}{2} + \cos\frac{3\pi x}{2} + \cdots \right]$$

$$a_n = (-1)^{n-1}\cos\left(\frac{n\pi x}{2}\right) \neq 0$$

series does not converges uniformly so, term by term different is not possible

Q.No8 Let $\{\phi_n(x)\}$ be a set mutually orthonormal function in (a,b). show that if $\sum_{n=1}^{\infty} c_n \phi_n(x)$ converges uniformally then

$$c_n = \int_a^b f(x)\,\phi_n(x)\,dx$$

Sol Series converges uniformaly to $f(x)$

$$f(x) = \sum_{n=1}^{\infty} c_n \phi_n(x)$$

Muliplying both sides $\phi_m(x)$

$$\int_a^b f(x)\,\phi_n(x)\,dx = \sum_{n=1}^{\infty} c_n \int_a^b \phi_n(x)\,\phi_m(x)\,dx$$

$$= \int_a^b \phi_m(x)\cdot\phi_n(x)\,dx \qquad \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

Hence

$$c_m = \int_a^b f(x)\,\phi_m(x)\,dx$$

and

$$c_n = \int_a^b f(x)\,\phi_n(x)\,dx.$$

Q.No9 Solve the integral by Fourier sine Transform.

$$\int_0^\infty f(x)\sin\alpha x\,dx = \begin{cases} 1-\alpha & 0\le\alpha\le 1 \\ 0 & \alpha>1 \end{cases}$$

Sol Given equation can be writen as

$$F_s[f(\alpha)] = F_s(\alpha) = \int_0^\infty F_s(x)\sin\alpha x\,dx$$

$$f(x) = \frac{2}{\pi}\int_0^1 (1-\alpha)\sin\alpha x\,dx$$

integrating and applying limits.

$$= \frac{2}{\pi}\left[(1-\alpha)\left(-\frac{\cos\alpha x}{x}\right)\Big|_0^1 + \int_0^1 \frac{\cos\alpha x}{x}(-1)\,d\alpha\right]$$

$$= \frac{2}{\pi}\left[0 - (-1)\left(\frac{-1}{x}\right) - \frac{1}{x}\left|\frac{\sin\alpha x}{x}\right|_0^1\right]$$

$$= \frac{2}{\pi}\left[\frac{1}{x} - \frac{\sin x}{x^2}\right]$$

$$f(x) = \frac{2}{\pi}\left(\frac{x-\sin x}{\quad}\right)$$